Project Code | *Kod Projek*	**BITU 3973**

UNIVERSITI TEKNIKAL MALAYSIA MELAKA

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

PROJEK SARJANA MUDA 1: PROPOSAL FORM

A TITLE OF PROPOSED PROJECT | *TAJUK PROJEK YANG DICADANGKAN*

EYEREMOTE : WEBCAM-BASED GAZE-CONTROLLED SYSTEM FOR ASSISTIVE TECHNOLOGY

B DETAILS OF STUDENT | *BUTIRAN PELAJAR*

Name	YAP ZHAN HONG			**Program:**	**BITC** ☐ **BITD** ☐ **BITI** ☐ **BITE** ☐ **BITM** ☐ **BITS** / **BITZ** ☐
Matric No.	B032310386				
Handphone No.	016-662 0589				
Semester	2/2	**Session**	2025/2026	**Email Address**	zhongyap154@gmail.com

C PROJECT INFORMATION | *MAKLUMAT PROJEK*

(i) Executive Summary of Project Proposal [Maximum 300 words]

Eye tracking technology has emerged as a vital assistive tool for individuals with physical disabilities, enabling them to control computers using only their eye movements. However, commercial eye tracking systems rely on expensive infrared hardware, making them unaffordable for many potential users. Recent advances in computer vision have made it possible to develop gaze tracking systems using standard webcams, significantly lowering the cost barrier. To address this issue, this project proposes the development of EyeRemote, a webcam-based gaze-controlled cursor system that enables hands-free computer interaction.

The primary problems identified include the high cost of existing eye tracking hardware, the lack of accessible solutions for low-income users, the absence of multi-user support in current webcam-based prototypes, and the limited interaction capabilities such as text selection and copy-paste functionality. This project aims to develop a comprehensive software solution that integrates real-time gaze estimation, cursor control, dwell-based clicking, blink detection for left and right clicks, multi-user face recognition for personalized calibration and security, and text manipulation features including selection and copy-paste operations. The system introduces a multi-user identification feature that ensures only authorized users can control the system and automatically loads their individual calibration settings, even when multiple faces appear in the camera view.

The expected outcomes include a fully functional webcam-based eye control system achieving two to three centimeters of screen accuracy, complete mouse emulation including left click, right click, double click, drag-and-drop, and scrolling, as well as text editing capabilities. By eliminating the need for specialized hardware, this system will significantly reduce the financial barrier to assistive technology and will be suitable for shared environments such as rehabilitation centers, special education schools, and family homes. Through the implementation of EyeRemote, this project aims to democratize access to gaze-based assistive technology and improve the quality of life for individuals with physical disabilities.

(ii) Detailed Proposal of the Project

(a) Introduction *(Project Background and Problem Statements)*

1. Project Background

Eye tracking technology has been widely used as an assistive tool for individuals with severe physical disabilities such as amyotrophic lateral sclerosis, cerebral palsy, spinal cord injury, and muscular dystrophy. These individuals often cannot use traditional input devices like mice or keyboards, making eye control their primary means of computer interaction. Commercial eye control systems enable users to perform all common computer tasks including browsing the web, typing, using social media, and running professional software with eye movement alone.

However, these solutions require specialized infrared eye trackers costing thousands of ringgits. Many potential beneficiaries, particularly In low-income communities and developing countries, cannot afford this assistive technology. Additionally, existing solutions are typically designed for single user and lack support for multiple users sharing the same device, which is common in rehabilitation centers, hospitals, and special education schools.

Recent advances in computer vision and machine learning have demonstrated the feasibility of accurate gaze estimation using standard webcams. Open-source framework such as MediaPipe Face Mesh provide real-time facial landmark detection. Recent research has shown that with appropriate calibration, webcam-based systems can achieve screen accuracy of two to three centimeters on large displays, which is sufficient for many assistive applications. Building upon these foundations, EyeRemote is proposed as a comprehensive, feature-rich webcam-based gaze-controlled system that not only replicated core functionality of traditional eye-tracking solution but also introduce multi-user support, advanced text manipulation, and intuitive shortcut controls.

2. Problem Statements

The high cost of commercial eye tracking solutions creates a significant accessibility barrier for individuals with physical disabilities who need hands-free computer control. Specialized eye tracking hardware remains prohibitively expensive for many potential users, particularly in low-income communities and developing regions. While recent advances computer vision have demonstrated the feasibility of gaze estimation using standard webcams, there is currently no freely available, fully-featured software solution that work with consumer-grade hardware.

Existing webcam-based prototypes lack essential features for practical deployment, particularly multi-user support. In shared environments such as rehabilitation centers, special education schools, and family homes, multiple users need to use the same system. Without user identification, personalized calibration settings cannot be loaded automatically, and when multiple people appear in the camera view, current system cannot determine who should control the cursor, leading to interference and confusion. There is also no protection against unauthorized access, which raised privacy and security concerns.

Furthermore, most low-cost prototypes offer only basic cursor control without comprehensive interaction capabilities. Advanced features such as text selection, copy-paste functionality, zoom control, right-click emulation, and intuitive shortcut panels are essential to perform real work independently. Without these features, webcam-based eye tacking is less practical as an assistive technology, leaving a gap between expensive commercial systems and simple research prototypes.

(b) Objectives of the Project

1. To develop a webcam-based gaze-controlled system that enable hands-free computer interaction through real-time gaze estimation and cursor control.
2. To design and implement interaction mechanisms that allows users to control computer functions using eye movements.
3. To create a user-friendly interface that support efficient and intuitive gaze-based operation.

(c) Scope of the Project

1. Module to be develop

a. User Management Module
This module handles user registration, profile management, and data storage. Users can create accounts, enroll their biometric information for recognition, and store personalized system setting and preferences. Multiple users can be registered under a single installation, with all personal data stored locally on the user's device to ensure privacy and security. The module provide a simple interface for adding, removing, and switching between users.

b. Face and Eye Landmark Detection Module
This module processes video input from a standard webcam to detect and track the user's face and eye regions in real time. It extracts relevant facial features with a focus on eye regions, including eye contours, corners, and pupil positions. The module is designed to handle variations in lighting conditions, different user appearances, and the presence of accessories such as glasses to ensure robust tracking performance. It may also measure additional physiological signals such as pupil size for advanced interaction features.

c. Gaze Estimation and Cursor Control Module
This module calculates the direction of the user's gaze based on the position of the eyes and other facial features. It compensates for natural head movements to maintain accurate tracking and maps the estimated gaze direction to screen coordinates using a mapping function provided by the calibration module. The module controls the movement of the on-screen cursor based on these coordinates and implements appropriate smoothing and filtering techniques to ensure comfortable and responsive cursor control.

d. User Identification and Access Control Module
This module performs real-time user recognition to identify authorized users of the system. When the system starts, it continuously analyzes the camera input to detect users and compares this information against enrolled user profiles. When a match is found with sufficient confidence, the system unlocks and loads that user's personalized settings automatically. When multiple users are detected simultaneously, the module identifies the primary user based on configurable priority rules and ensures that only the active user's gaze is used for system control. If no authorized users remain in view, the system locks to prevent unauthorized access.

e. Calibration Module
This module establishes the mathematical relationship between the user's gaze characteristics and actual screen coordinates. It includes an explicit calibration procedure where users follow visual targets on the screen to collect reference data. This data is used to create a personalized mapping function that adapts the system to each user's unique eye characteristics and viewing setup. The module may also support ongoing refinement by collecting additional data during normal system use to gradually improve accuracy and adapt to environmental changes.

f. Interaction Recognition Module
This module interprets user intent by analyzing gaze behavior and other eye-related signals in real time. It detects intentional dwell-based selections when the user's gaze remains within a defined screen region for a specified duration. The module recognizes different patterns of eye gestures such as blinks, winks, or other voluntary eye movements, and maps them to various commands including mouse clicks, mode switching, and function activation. It also detects when the user's gaze approaches screen boundaries to trigger scrolling actions. For advanced functions such as zoom control, the module may interpret sustained gaze on interface elements or analyze physiological signals. The module includes appropriate filtering mechanisms to distinguish intentional user actions from natural eye behavior.

g. User Guidance and Setup Module
This module assists users with proper positioning and initial system setup. It provides visual feedback to help users position themselves optimally relative to the camera for accurate tracking. Visual indicators guide users to adjust their distance, position, or orientation until optimal conditions are achieved. Once proper positioning is confirmed, the module guides users through the calibration process with clear on-screen instructions and visual cues. After calibration, it provides verification tools for users to confirm that the system is working correctly.

h. Text Input and Manipulation Module
This module provides an on-screen interface for text entry optimized for gaze-based interaction.

It includes a virtual keyboard with appropriate visual design to facilitate accurate selection using gaze. The module supports essential text input features such as character entry, word prediction, and language switching. Beyond basic typing, this module handles text editing operations including selection of text ranges, copy and paste functionality, text deletion, and other common editing tasks. Selected text is visually highlighted for confirmation. The module may also integrate optical character recognition capabilities to allow users to capture and manipulate text from non-editable areas of the screen such as images or application dialogs.

I. Notification and Feedback Module

This module keeps users informed about system status through appropriate audio and visual channels. It provides notifications for important events such as successful user recognition, completion of calibration, detection of multiple users, system lock and unlock events, and any errors that may occur during operation. All notifications are designed to be noticeable without being intrusive, with options to adjust feedback modalities based on user preferences.

j. User Interface and Control Panel Module

This module provides a user interface optimized for gaze-based interaction. It features a flexible control panel with clearly labeled and appropriately sized controls for functions such as mouse mode selection, text input, view navigation, and system settings. The interface offers real-time visual feedback, displaying system status, detected user information, and gaze position. A dedicated settings section allows users to configure parameters such as sensitivity, feedback preferences, and user management. All interface elements are designed with gaze interaction in mind, ensuring suitable target sizes and clear visual cues for intuitive operation.

2. Target User

a. Individuals with physical disabilities

The primary target users are individuals with physical disabilities including those with amyotrophic lateral sclerosis, cerebral palsy, spinal cord injury, muscular dystrophy, stroke survivors, and others with limited or no hand use. These users will benefit from the ability to control computers independently using only their eyes.

b. Rehabilitation centers and special education schools

Secondary users include rehabilitation centers and special education schools that need low-cost, deployable gaze interaction tools for therapy, communication, and education purposes. The multi-user support makes the system particularly valuable in these shared environments.

c. Caregivers and Family Members

Caregivers and family members are also target users as they will need to set up and manage the system for their loved ones. The intuitive interface and clear guidance modules are designed to make this possible even for non-technical users.

d. Researchers

Researchers in human-computer interaction and assistive technology may use the system as a platform for conducting gaze behavior studies and user experiments with accessible hardware.

e. Enthusiasts and Developers

General enthusiasts and developers interested in hands-free computing or assistive technology may use the system as a foundation for their own projects, particularly given the open-source nature of the planned release.

(d) Expected Outcome/ Proposed Solution

1) Increases accessibility to assistive technology

EyeRemote is designed to make gaze-controlled computing accessible to anyone with a standard computer and webcam. By eliminating the need for expensive infrared hardware, the system reduces the cost barrier from thousands of ringgits to zero. This democratization of assistive technology will benefit individuals with disabilities who cannot afford commercial solutions, as well as institutions in developing regions with limited budgets.

2) Provides multi-user support for shared environments

Unlike existing webcam-based prototypes that support only single users, EyeRemote introduces comprehensive multi-user face recognition and locking. The system automatically identifies authorized users and loads their personalized calibration settings. When multiple people appear in the camera view, the system intelligently selects the primary user and ignores others, preventing interference. This makes the system practical for rehabilitation centers, special education schools, hospitals, and family homes where multiple users share the same computer.

3) Ensures privacy and security through face locking

The face recognition module ensures that only authorized users can control the system. When an unrecognized face appears, the system locks, preventing unauthorized access to the computer. All face templates are stored locally, protecting user privacy. This feature gives caregivers peace of mind that the system will not be misused by children or visitors.

4) Enhances accuracy through adaptive calibration

The adaptive calibration module combines explicit calibration with continuous implicit refinement. Users perform a simple 5-point calibration during initial setup, and the system continues to learn from natural interactions, adapting to changes in lighting, distance, and head position. This reduces the need for frequent manual recalibration and maintains accuracy over time.

5) Provides complete hands-free interaction

Beyond basic cursor control, EyeRemote includes dwell-based clicking, blink pattern recognition for left and right clicks, edge-triggered scrolling, and an on-screen keyboard for text input. These features enable users to perform all common computer tasks without any physical input devices, achieving true hands-free operation.

6) Offers intuitive setup guidance

The face alignment frame with real-time color feedback guides first-time users through proper positioning. The step-by-step calibration process with moving targets is easy to follow even for non-technical users. This reduces the need for technical assistance and empowers users to set up the system independently.

D REFERENCES | *RUJUKAN*

State your references (Minimum 10 references)

1	Fenyom, I., Adedigba, A., Darlan, D. et al. (2026). Lightweight webcam-based eye tracking system for large display screens. Multimedia Tools and Applications, 85. https://link.springer.com/article/10.1007/s11042-025-15678-2
2	Blakey, W., Katsigiannis, S., & Ramzan, N. (2025). A study on the impact of different components of a traditional webcam-based 2D gaze tracking algorithm. IEEE Sensors Journal. https://ieeexplore.ieee.org/document/10895672

3	Hasnul, M. A., & Aziz, N. A. A. (2025). Robust Camera-Based Eye-Tracking Method Allowing Head Movements and Its Application in User Experience Research. Journal of Eye Movement Research, 18(2). https://bop.unibe.ch/JEMR/article/view/11234
4	Puspasari, M. A. (2025). Multi-Eyes: Multi-User Eye Tracking Using Commodity Hardware. Doctoral dissertation, Old Dominion University. https://digitalcommons.odu.edu/computerscience_etds/156/
5	Kishi, N., & Hayashi, M. (2015). Effective Gaze Writing with Support of Text Copy and Paste. 2015 IEEE/ACIS 14th International Conference on Computer and Information Science. https://ieeexplore.ieee.org/document/7166589
6	Soukupová, T., & Čech, J. (2016). Real-time eye blink detection using facial landmarks. 21st Computer Vision Winter Workshop. https://www.researchgate.net/publication/303572073_Real-time_eye_blink_detection_using_facial_landmarks
7	Krafka, K., et al. (2016). Eye tracking for everyone. IEEE Conference on Computer Vision and Pattern Recognition. https://www.cv-foundation.org/openaccess/content_cvpr_2016/html/Krafka_Eye_Tracking_for_CVPR_2016_paper.html
8	Papoutsaki, A., et al. (2016). WebGazer: Scalable webcam eye tracking using user interactions. International Joint Conference on Artificial Intelligence. https://www.ijcai.org/Proceedings/16/Papers/545.pdf
9	MediaPipe Face Mesh. Google. https://developers.google.com/mediapipe/solutions/vision/face_mesh
10	OpenCV Library https://opencv.org/

E	**DECLARATION BY STUDENT** | *AKUAN PELAJAR*	
(i)	**Date:** 6/3/2026	**Student's Signature:** hong

E	**RECOMMENDED BY SUPERVISOR** *PERAKUAN OLEH PENYELIA*
(ii)	**Recommended** ☐ **Not Recommended** ☐
	Comments:
	Supervisor's Name:
	Signature & Stamp:
	Date: